

Technical Formulation and Parameter Justification for the Variable-Compliance Rover Mobility Model

Jane Park

September 7, 2026

1 Purpose

This document records the mathematical formulation used in the variable-compliance rover mobility model and identifies the justification and provenance of each term. It is intended to accompany the research report and source code.

2 State variable

The wheel state is represented by

$$s \in [0, 1]. \quad (1)$$

Recovered source anchor states are

$$s_k = \{0, 0.149537, 0.422427, 0.572765, 0.746899, 0.916081, 1\}. \quad (2)$$

The continuous constitutive response is

$$D = D(F, s), \quad (3)$$

constructed from Lee et al. force–deformation data [1].

Justification. A continuous state avoids imposing an artificial binary soft/stiff wheel and permits optimization over the experimentally represented compliance range.

3 Normal load

For the present single-wheel-equivalent calibration,

$$F_N = W = mg. \quad (4)$$

Using

$$m = 5.3 \text{ kg}, \quad g = 9.81 \text{ m s}^{-2}, \quad (5)$$

gives

$$F_N = 51.993 \text{ N}. \quad (6)$$

For a general N_w -wheel rover under a static equal-load approximation,

$$F_{N,k} \approx \frac{m_{\text{rover}} g}{N_w}. \quad (7)$$

Justification. The current fixed value preserves consistency with the Lee small-wheel data. A general rover implementation should use per-wheel load rather than total rover weight.

4 Wheel geometry

The current calibration uses

$$R = 0.140 \text{ m}, \quad b_w = 0.040 \text{ m}. \quad (8)$$

Justification. These dimensions are tied to the source wheel configuration used to define $D(F, s)$. They are not yet free wheel-family parameters.

5 Contact geometry

For radial deformation D ,

$$L_c = 2\sqrt{2RD - D^2}. \quad (9)$$

The reduced contact area is

$$A = b_w L_c, \quad (10)$$

and the effective Bekker width is

$$b = \min(b_w, L_c). \quad (11)$$

Average pressure is

$$p = \frac{F_N}{A}. \quad (12)$$

Justification. This reduced geometry allows the Lee-derived deformation response to modify the contact patch used by the terramechanics calculation. It is an approximation and does not represent the full pressure distribution.

6 Pressure–sinkage relation

For deformable terrain,

$$\sigma = \left(\frac{k_c}{b} + k_\phi \right) y^n, \quad (13)$$

following classical Bekker/Wong terramechanics [2, 4].

Using the average pressure p ,

$$z = \left[\frac{p}{k_c/b + k_\phi} \right]^{1/n}. \quad (14)$$

Parameters. k_c , k_ϕ , and n are terrain-specific pressure–sinkage parameters.

Justification. These parameters are terrain properties rather than wheel fitting coefficients. They must be traced to their literature or inherited data source.

7 Compaction resistance

The reduced compaction resistance is

$$R_c = b_w \left(\frac{k_c}{b} + k_\phi \right) \frac{z^{n+1}}{n+1}. \quad (15)$$

Justification. This term is a reduced work-per-distance representation of soil compaction. It provides a state-dependent mobility cost because compliance changes contact pressure and sinkage.

8 Traction feasibility

Available traction is bounded by

$$F_{\max} = cA + F_N \tan \phi, \quad (16)$$

where c is cohesion and ϕ is internal friction angle.

The state is rejected if

$$F_{\text{req}} > F_{\max}. \quad (17)$$

Justification. This is used only as a feasibility bound. It is not interpreted as a calibrated slip prediction. A Janosi–Hanamoto slip-energy model was not added because additional wheel-specific calibration would be required.

9 Slope

For a directed graph edge $i \rightarrow j$ with physical length L_{ij} ,

$$\theta_{ij} = \tan^{-1} \left(\frac{z_j - z_i}{L_{ij}} \right). \quad (18)$$

The gravitational contribution is

$$F_g = W \sin \theta_{ij}. \quad (19)$$

Justification. Signed slope is computed directly from elevation and physical edge length. An uphill regression reproduced the expected gravitational energy increment $W\Delta z$.

10 Positive-propulsion objective

The raw mobility demand is converted to positive propulsion demand as

$$F_{\text{prop}} = \max(0, F_{\text{raw}}). \quad (20)$$

Justification. The model does not assign regenerative credit or braking losses. Therefore downhill results must be interpreted as positive propulsion demand only.

11 Rolling-loss sensitivity term

The reduced state-dependent rolling term is

$$R_{\text{roll}}(s) = W [C_{rr,\text{ref}} + \lambda h(s)]. \quad (21)$$

The model retains four alternative $h(s)$ forms:

- (1) linear state,
- (2) square-root loading work,
- (3) loading work,
- (4) deflection.

The tested sensitivity set uses

$$C_{rr,\text{ref}} \in \{0.0005, 0.001, 0.002, 0.005\}, \quad (22)$$

and

$$\lambda \in \{0.003, 0.01, 0.02\}. \quad (23)$$

Justification. Lee’s quasi-static deformation data do not calibrate cyclic rolling dissipation. The rolling term is therefore treated as model-form and amplitude sensitivity rather than a measured law.

12 Terrain cost density

For each terrain T , slope θ , and wheel state s , the model evaluates

$$c(T, \theta, s) \quad (24)$$

in units of positive propulsion work per distance.

For deformable terrain this includes the reduced soil-compaction term, rolling term, signed gravitational contribution, and traction feasibility.

13 Graph edge length

For physical grid spacing Δx ,

$$L_{ij} = \begin{cases} \Delta x, & \text{cardinal edge,} \\ \sqrt{2}\Delta x, & \text{diagonal edge.} \end{cases} \quad (25)$$

Justification. Δx is the spatial resolution of the map. It is not fitted. A refinement audit demonstrated first-order convergence of adaptive terrain-interface error.

14 Edge cost

For an edge connecting terrain endpoints i and j ,

$$C_{ij}(s) = L_{ij} \frac{c_i(s, \theta_{ij}) + c_j(s, \theta_{ij})}{2}. \quad (26)$$

Justification. Endpoint averaging replaces destination-only cost assignment and reduces directional bookkeeping bias at terrain boundaries.

15 Grid connectivity

The final 2-D formulation uses 8-neighbor connectivity.

Justification. A 4-neighbor grid produced a resolution-independent 41.42% path-length error for 45° travel. The 8-neighbor formulation removes that error at 45° . Residual maximum angular path-length error is approximately 8.24% near 22.5° .

16 Obstacle feasibility

For an explicitly defined directed obstacle edge of tested height h ,

$$s_e \geq s_{\min}(h). \quad (27)$$

Justification. Only exact Lee-tested heights are supported. The obstacle data define passability, not energy.

17 Fixed optimization

For a fixed state s and path P ,

$$J(P, s) = \sum_{e \in P} C_e(s). \quad (28)$$

The globally optimal fixed solution is

$$J_{\text{fixed}}^* = \min_s \min_P J(P, s). \quad (29)$$

Justification. This establishes a strong baseline. The adaptive wheel is not compared against an arbitrarily selected fixed compliance.

18 Adaptive optimization

For an adaptive state sequence $\{s_e\}$,

$$J_{\text{adaptive}}^* = \min_{P, \{s_e\}} \sum_{e \in P} C_e(s_e). \quad (30)$$

Transition energy is currently

$$E_{\text{switch}} = 0. \quad (31)$$

Justification. Actuation energy and transition distance are not sufficiently known. Setting transition energy to zero deliberately defines an ideal mobility benefit rather than inventing an unsupported penalty.

19 Ideal mobility benefit

Define

$$\rho = \frac{J_{\text{adaptive}}^*}{J_{\text{fixed}}^*}. \quad (32)$$

Then

$$B_{\text{ideal}} = 1 - \rho = \frac{J_{\text{fixed}}^* - J_{\text{adaptive}}^*}{J_{\text{fixed}}^*}. \quad (33)$$

Since the adaptive optimizer can reproduce any fixed state,

$$J_{\text{adaptive}}^* \leq J_{\text{fixed}}^*, \quad (34)$$

and therefore

$$B_{\text{ideal}} \geq 0. \quad (35)$$

20 Benefit decomposition

Let $J_{a|f}$ denote adaptive state selection constrained to the best fixed route. Then

$$\Delta J_{\text{state}} = J_f - J_{a|f}, \quad (36)$$

$$\Delta J_{\text{route}} = J_{a|f} - J_a, \quad (37)$$

and

$$\Delta J_{\text{total}} = \Delta J_{\text{state}} + \Delta J_{\text{route}}. \quad (38)$$

Justification. This separates savings caused by changing wheel state from savings caused by selecting a different route.

21 Parameter/provenance table

Quantity	Current value/range	Classification	Justification / status
m	5.3 kg	source-matched nominal	Lee small-wheel anchored; not universal rover mass.
F_N	51.993 N	derived	mg for current calibration; general rover should use per-wheel load.
R	0.140 m	source geometry	Not yet a free wheel-family parameter.
b_w	0.040 m	source geometry	Same limitation as R .
s	0–1	source-derived	Recovered continuous compliance coordinate.
$D(F, s)$	recovered surface	source-derived	Main wheel constitutive input.
k_c, k_ϕ, n	terrain-specific	literature/inherited	Pressure–sinkage properties.
c, ϕ	terrain-specific	literature/inherited	Traction-feasibility properties.
$C_{rr, \text{ref}}$	swept	sensitivity	Not a measured Lee-wheel value.

λ	swept	sensitivity	State-dependent rolling-loss amplitude.
$h(s)$	four forms	model-form sensitivity	Explicit uncertainty treatment.
Δx	map-dependent	physical/numerical	Physical map resolution; convergence audited.
Connectivity	8-neighbor	numerical choice	Selected after quantified Manhattan bias.
E_{switch}	0	idealization	Transition cost unknown.
Obstacle energy	none	excluded	Data support feasibility only.
Downhill regen	none	reduced objective	Negative propulsion clipped to zero.

22 Generality boundary

The graph and optimization formulation can accept different wheel loads, geometries, terrain maps, and route configurations in principle. The current physical calibration cannot yet support arbitrary wheel geometry.

A broader wheel-family model requires either:

- (1) experimental data at multiple wheel sizes and loads;
- (2) validated FEA across geometry/load space; or
- (3) a mechanics-based dimensionless scaling law.

Until such validation is available, changes in R or b_w should not be interpreted as validated predictions.

References

- [1] J.-Y. Lee et al., “Variable-stiffness–morphing wheel inspired by the surface tension of a liquid droplet,” *Science Robotics*, vol. 9, no. 93, ead12067, 2024. doi: 10.1126/scirobotics.adl2067.
- [2] M. G. Bekker, *Introduction to Terrain-Vehicle Systems*. University of Michigan Press, 1969.
- [3] J. Y. Wong, *Theory of Ground Vehicles*, 3rd ed. John Wiley & Sons, 2001.
- [4] J. Y. Wong, *Terramechanics and Off-Road Vehicle Engineering: Terrain Behaviour, Off-Road Vehicle Performance and Design*, 2nd ed. Elsevier, 2010.
- [5] Y. Shen, M. Zou, H. Cao, D. Pan, B. Yuan, and L. He, “Current-Based Analysis and Validation of a Wheel–Soil Interaction Model for the Trafficability of a Planetary Rover,” *Aerospace*, vol. 11, no. 11, p. 892, 2024. doi: 10.3390/aerospace11110892.